

## Semiconductor packaging industry update

EQUITY: JAPAN ELECTRONIC PARTS

## Packaging substrate shipments up 20% y-y

Start of competition for supremacy in 3.5D packaging

**Packaging-related monthly statistics: April shipments down from March record high**

According to the *Current Survey of Production* for April, released on 12 June, shipments of packaging substrates (rigid module substrates) rose 20% y-y to ¥23.3bn, down from ¥27.3bn in March. Shipments in terms of m<sup>2</sup> rose 1% y-y and the average price per m<sup>2</sup> rose 19% y-y, and only slightly m-m, to a record high of ¥1.226mn. Based on conditions in other supply chains, we think shipments of packaging for Rubin processors may have been temporarily sluggish because of the pushing back of server production. Shipments in April 2025, when Blackwell packages got into full swing, were high in m<sup>2</sup> terms. We think that shipments of Rubin packages will move into full swing in summer 2026 (June at the earliest and around August at the latest).

**Adoption of 3.5D packaging likely to pick up in 2028: Moves at TSMC, Intel, Google**

In the packaging industry, there has been an increase in news related to 3.5D packaging, mass production of which is due to start in 2028. 3.5D packaging is a back-end process technology that involves hybrid bonding of accelerators, HBM, and small inference processors, and bonding them together on the package. At its AGM, TSMC said that: (1) it plans to launch a pilot production line for panel-level packaging (Chip-on-Panel-on-Substrate (CoPoS)) in June 2026 and start mass production in 2028 following verification work; and (2) it plans to prepare for mass production of glass core substrates with partners (Ibiden, etc) over the next two to three years, despite issues in development. Based on these comments, we think that TSMC's 3.5D packaging consists of chip stacking using System on Integrated Chips (SoIC), CoPoS, and Flip Chip Ball Grid Array (FC-BGA) using glass core substrates (however, adoption of glass core substrates is likely to come after that of CoPoS). It appears that Intel's Embedded Multi-die Interconnect Bridge Through-Silicon Via (EMIB-T) technology is being adopted in 3.5D packaging and that companies are increasingly making moves to place more orders for it, perhaps because of uncertainty about when TSMC's CoPoS will go into mass production. For example, *The Information* reported on 8 June that Google/Alphabet had placed orders for Intel to package at least 3mn TPUs with EMIB-T in 2028, and that Nvidia was considering adopting the technology (we have not confirmed the accuracy of the report). As Ibiden does not have the capacity to supply substrates for the packaging of 3mn TPUs per year with EMIB-T technology, we do not expect the news to be immediately reflected in Ibiden's earnings forecasts, but we will be interested to see if Ibiden announces any plans to increase production capacity.

One point to watch with regard to the adoption of 3.5D packaging is that it won't be easy for companies developing accelerators on TSMC's back-end process design platform (TSMC 3DFabric Alliance) to transition to EMIB-T, as 3.5D packaging includes key functions such as power supply and heat management. EMIB-T design will require companies to either rely on Intel's back-end process design platform or to build their own back-end semiconductor process supply chains. In power supply, we think vertical power supply or integrated voltage regulators (iVRs) would have to be developed jointly with partners, while in heat management, new cold plates with microchannel mechanisms optimized for cooling each chip would require the same. We expect collaboration among companies to pick up.

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Japan electronic parts

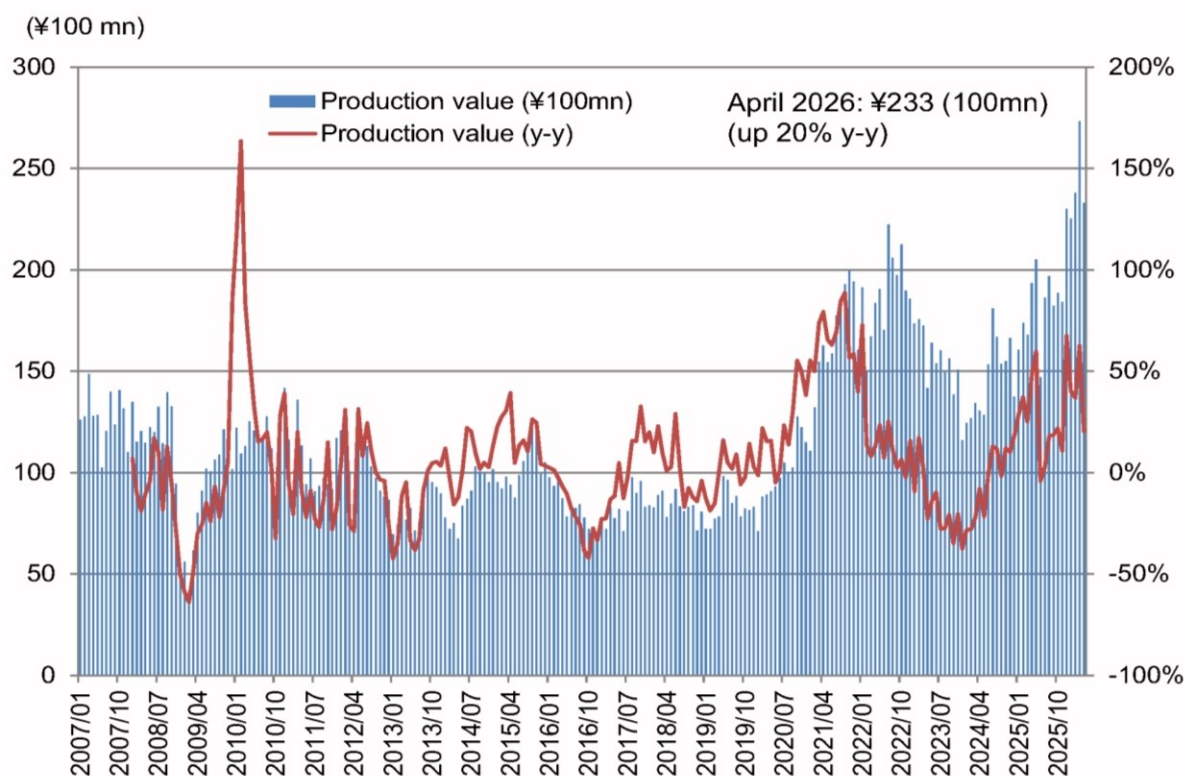
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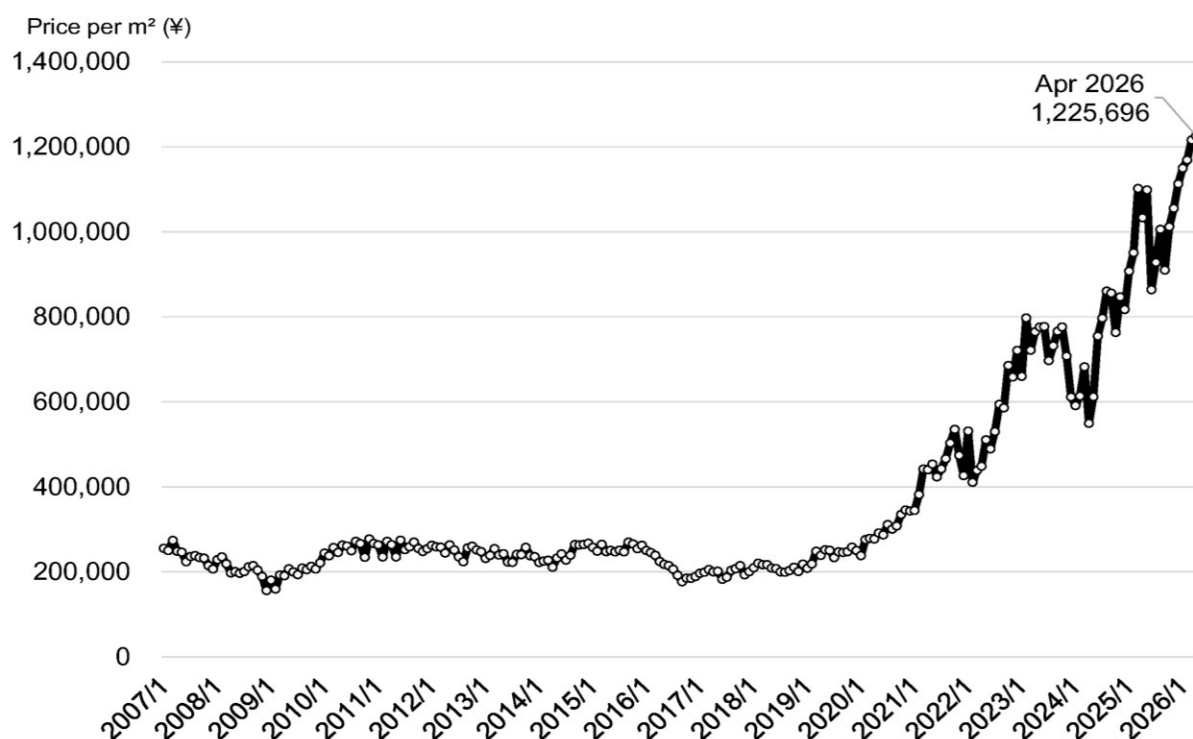
# Reference figures

**Fig. 1: Rigid PCB production value**



Source: Nomura, based on Ministry of Economy, Trade & Industry's Current Survey of Production

**Fig. 2: Price per m<sup>2</sup> for rigid module boards**



Note: Module boards have components mounted on the board surface.

Source: Nomura, based on Ministry of Economy, Trade & Industry's Current Survey of Production



# Appendix A-1

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In addition, all products carry the risk of losses owing to price fluctuations or other factors. Fees and risks vary by product. Please thoroughly read the written materials provided, such as documents delivered before making a contract, listed securities documents, or prospectuses. Transactions involving Japanese equities (including Japanese REITs, Japanese ETFs, and Japanese ETNs, Japanese Infrastructure Funds) are subject to a sales commission of up to 1.43% (tax included) of the transaction amount (or a commission of ¥2,860 (tax included) for transactions of ¥200,000 or less). When Japanese equities are purchased via OTC transactions (including offerings), only the purchase price shall be paid, with no sales commission charged. However, Nomura Securities may charge a separate fee for OTC transactions, as agreed with the customer. Japanese equities carry the risk of losses owing to price fluctuations. Japanese REITs carry the risk of losses owing to fluctuations in price and/or earnings of underlying real estate. Japanese ETFs and ETNs carry the risk of losses owing to fluctuations in the underlying indexes or other benchmarks. Japanese Infrastructure Funds carry out the risk of losses owing to fluctuations in price and/or earnings of underlying infrastructures.

Transactions involving foreign equities are subject to a domestic sales commission of up to 1.045% (tax included) of the transaction amount (which equals the local transaction amount plus local fees and taxes in the case of a purchase or the local transaction amount minus local fees and taxes in the case of a sale) (for transaction amounts of ¥750,000 and below, maximum domestic sales commission is ¥7,810 (tax included)). Local fees and taxes in foreign financial instruments markets vary by country/territory. When foreign equities are purchased via OTC transactions (including offerings), only the purchase price shall be paid, with no sales commission charged. However, Nomura Securities may charge a separate fee for OTC transactions, as agreed with the customer. Foreign equities carry the risk of losses owing to factors such as price fluctuations and foreign exchange rate fluctuations.

Margin transactions are subject to a sales commission of up to 1.43% (tax included) of the transaction amount (or a commission of ¥2,860 (tax included) for transactions of ¥200,000 or less), as well as management fees and rights handling fees. In addition, long margin transactions are subject to interest on the purchase amount, while short margin transactions are subject to fees for the lending of the shares borrowed. A margin equal to at least 30% of the transaction amount (at least 33% for online transactions) and at least ¥300,000 is required. With margin transactions, an amount up to roughly 3.3x the margin (roughly 3x for online transactions) may be traded. Margin transactions therefore carry the risk of losses in excess of the margin owing to share price fluctuations. For details, please thoroughly read the written materials provided, such as listed securities documents or documents delivered before making a contract.

Transactions involving convertible bonds are subject to a sales commission of up to 1.10% (tax included) of the transaction amount (or a commission of ¥4,400 (tax included) if this would be less than ¥4,400). When convertible bonds are purchased via OTC transactions (including offerings), only the purchase price shall be paid, with no sales commission charged. However, Nomura Securities may charge a separate fee for OTC transactions, as agreed with the customer. Convertible bonds carry the risk of losses owing to factors such as interest rate fluctuations and price fluctuations in the underlying stock. In addition, convertible bonds denominated in foreign currencies also carry the risk of losses owing to factors such as foreign exchange rate fluctuations.

When bonds are purchased via public offerings, secondary distributions, or other OTC transactions with Nomura Securities, only the purchase price shall be paid, with no sales commission charged. Bonds carry the risk of losses, as prices fluctuate in line with changes in market interest rates. Bond prices may also fall below the invested principal as a result of such factors as changes in the management and financial circumstances of the issuer, or changes in third-party valuations of the bond in question. In addition, foreign currency-denominated bonds also carry the risk of losses owing to factors such as foreign exchange rate fluctuations.

When Japanese government bonds (JGBs) for individual investors are purchased via public offerings, only the purchase price shall be paid, with no sales commission charged. As a rule, JGBs for individual investors may not be sold in the first 12 months after issuance. When JGBs for individual investors are sold before maturity, an amount calculated via the following formula will be subtracted from the par value of the bond plus accrued interest: (1) for 10-year variable rate bonds, an amount equal to the two preceding coupon payments (before tax) x 0.79685 will be used, (2) for 5-year and 3-year fixed rate bonds, an amount equal to the two preceding coupon payments (before tax) x 0.79685 will be used. When inflation-indexed JGBs are purchased via public offerings, secondary distributions (uridashi deals), or other OTC transactions with Nomura Securities, only the purchase price shall be paid, with no sales commission charged. Inflation-indexed JGBs carry the risk of losses, as prices fluctuate in line with changes in market interest rates and fluctuations in the nationwide consumer price index. The notional principal of inflation-indexed JGBs changes in line with the rate of change in nationwide CPI inflation from the time of its issuance. The amount of the coupon payment is calculated by multiplying the coupon rate by the notional principal at the time of payment. The maturity value is the amount of the notional principal when the issue becomes due. For J117 and subsequent issues, the maturity value shall not undercut the face amount. Purchases of investment trusts (and sales of some investment trusts) are subject to a purchase or sales fee of up to 5.5% (tax included) of the transaction amount. Also, a direct cost that may be incurred when selling investment trusts is a fee of up to 2.0% of the unit price at the time of redemption. Indirect costs that may be incurred during the course of holding investment trusts include, for domestic investment trusts, an asset



management fee (trust fee) of up to 5.5% (tax included/annualized basis) of the net assets in trust, as well as fees based on investment performance. Other indirect costs may also be incurred. For foreign investment trusts, indirect fees may be incurred during the course of holding such as investment company compensation.

Investment trusts invest mainly in securities such as Japanese and foreign equities and bonds, whose prices fluctuate. Investment trust unit prices fluctuate owing to price fluctuations in the underlying assets and to foreign exchange rate fluctuations. As such, investment trusts carry the risk of losses. Fees and risks vary by investment trust. Maximum applicable fees are subject to change; please thoroughly read the written materials provided, such as prospectuses or documents delivered before making a contract.

In interest rate swap transactions and USD/JPY basis swap transactions ("interest rate swap transactions, etc."), only the agreed transaction payments shall be made on the settlement dates. Some interest rate swap transactions, etc. may require pledging of margin collateral. In some of these cases, transaction payments may exceed the amount of collateral. There shall be no advance notification of required collateral value or collateral ratios as they vary depending on the transaction. Interest rate swap transactions, etc. carry the risk of losses owing to fluctuations in market prices in the interest rate, currency and other markets, as well as reference indices. Losses incurred as such may exceed the value of margin collateral, in which case margin calls may be triggered. In the event that both parties agree to enter a replacement (or termination) transaction, the interest rates received (paid) under the new arrangement may differ from those in the original arrangement, even if terms other than the interest rates are identical to those in the original transaction. Risks vary by transaction. Please thoroughly read the written materials provided, such as documents delivered before making a contract and disclosure statements.

In OTC transactions of credit default swaps (CDS), no sales commission will be charged. When entering into CDS transactions, the protection buyer will be required to pledge or entrust an agreed amount of margin collateral. In some of these cases, the transaction payments may exceed the amount of margin collateral. There shall be no advance notification of required collateral value or collateral ratios as they vary depending on the financial position of the protection buyer. CDS transactions carry the risk of losses owing to changes in the credit position of some or all of the referenced entities, and/or fluctuations of the interest rate market. The amount the protection buyer receives in the event that the CDS is triggered by a credit event may undercut the total amount of premiums that he/she has paid in the course of the transaction. Similarly, the amount the protection seller pays in the event of a credit event may exceed the total amount of premiums that he/she has received in the transaction. All other conditions being equal, the amount of premiums that the protection buyer pays and that received by the protection seller shall differ. In principle, CDS transactions will be limited to financial instruments business operators and qualified institutional investors. Transfers of equities to another securities company via the Japan Securities Depository Center are subject to a transfer fee of up to ¥11,000 (tax included) per issue transferred depending on volume. No account fee will be charged for marketable securities or monies deposited.

## **Nomura Securities Co., Ltd.**

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